

The Categorical Structure of Federated Convergence: Limits, Sheaves, and Minimal Coordination

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Abstract

Normalization confluence gives coordination-free convergence within a single registry, and its federated extension gives convergence across a network of registries connected by morphisms, under two decidable regimes: acyclic networks, and monotone repair on cycles [1, 2]. Those regimes are stated as separate sufficient conditions. This paper gives the structure underneath them and, with it, an answer to a sharper question: when must a federation coordinate at all, and how little coordination suffices?

We show that a federation’s convergent states are a limit and its federated normalizer is the retraction onto that limit, so compositionality (a verified sub-federation collapses to an effective registry) is a corollary of one structural fact rather than a per-topology argument. We then show that convergence certificates form a sheaf whose gluing condition is exactly the source-determinacy and validity-preservation hypotheses the federation already certifies, so the obstruction to a global convergent state is cohomological: H^0 is the convergent states and H^1 is the holonomy of the morphism cycles. The two convergence regimes are then one picture: an acyclic network has no cycles and H^1 vanishes, while a monotone cycle collapses the obstruction to a least fixed point. The obstruction is computable as a loop composite over the shared subspace, implemented in the reference engine as a cycle diagnostic that names an obstructing cycle and exhibits its orbit. Finally we show that when the obstruction is nonzero, coordinating a cycle basis of the morphism network suffices to converge, and in the information-preserving (invertible) fragment this coordinated core is minimal: a first-Betti-number many loops, localizing the global coordination verdict of CALM and invariant confluence into a minimal, mixed-consistency partition. The structural core (the limit, the retraction, and compositionality) is mechanized axiom-free in Coq/Rocq.

Keywords: Federated Convergence, Normalization Confluence, Sheaf Cohomology, Coordination Avoidance, CALM, Category Theory, Limits, Holonomy, Minimal Coordination, Mechanized Proof

1 Introduction

Normalization confluence certifies that a single registry converges: under well-founded compensation and compensation commutativity, every processor that consumes the same events reaches the same valid state regardless of order [1]. Its federated extension certifies that a network of such registries, connected by directed morphisms that carry one registry’s authority over another’s shared state, converges as a whole, in two regimes: any acyclic network whose morphisms preserve validity, and any network (cyclic included) whose repair is monotone [2]. Both regimes are established by finite enumeration and are what the reference engine checks at build time.

Those two regimes are presented as independent sufficient conditions, proved by separate arguments. That leaves a structural question unanswered and a practical one unasked. The structural question: is there one reason both regimes work, rather than two? The practical question: when a federation falls into neither regime, the engine rejects it, but rejection is a blunt verdict. A designer wants to know not merely that the network cannot converge coordination-free, but *where* the conflict is and *how little* coordination would repair it. This paper answers both, and the answers turn out to be the same object viewed twice.

The local-to-global view. A federation poses a local-to-global problem. Each registry is convergent on its own; each morphism, in isolation, is a clean rule for propagating authority. The question is whether, once every morphism constraint holds at once, there is a single global assignment the whole network agrees on, and whether it is reached regardless of event order. This is the shape of a limit, and of a sheaf gluing problem, and the obstruction to it is the shape of a cohomology class. Making that precise is what unifies the regimes: an acyclic network has no cycles to obstruct gluing, and a monotone cycle has an order that forces gluing even when a cycle is present. The same H^1 that measures the obstruction also, when it is nonzero, pinpoints the cycles responsible, which is exactly the diagnostic and the minimal-coordination answers the practical question wants.

Scope, stated up front. The cleanest form of the cohomological classification holds in the *invertible* fragment, where each morphism transports its shared value by a bijection (a lossless relabeling: a copy, a permutation, a bijective unit or schema map). There the transitions are group elements, H^1 is genuine Čech cohomology, and the minimal coordinated core is a cycle basis of exactly the first Betti number. When morphisms collapse information (a clamp, a reset, a most-restrictive-wins merge, the shape most compensation takes), the transitions are monoid elements rather than group elements, classical H^1 does not apply, and the correct statement is the operational one: a global convergent state exists iff the loop composite of each cycle has a reachable fixed point. We therefore lead with the general, operational results (the limit, the retraction, compositionality, and the loop-composite diagnostic), which hold for arbitrary morphisms and are what the engine implements, and present the cohomology as the sharp closed form on the invertible fragment. We state which real federations inhabit that fragment (Section 9) rather than leaving it as “the case where the math is clean.”

Contributions.

1. **Federation as a limit, and compositionality for free (Sections 4–5).** A registry’s valid set is the equalizer of its idempotent normalizer, a federation’s convergent set is the equalizer of its shared-projection and resolver maps, and the federated normalizer is the

idempotent retraction onto that limit. Because the normal forms are a limit, a convergent sub-federation collapses to an effective registry and the combined normalizer factors through it; compositionality is a corollary of one structural fact, not a per-topology theorem.

2. **Convergence certificates form a sheaf, and $R1/R2$ is its gluing axiom (Section 6).** Local convergence certificates glue to a global one exactly when the subsystems agree as normalizers on their overlap. The two regimes that force agreement, single-writer authority and monotone overlap, are the federation’s source-determinacy ($R1$) and validity-preservation ($R2$) hypotheses. The existing federation design is validated as the sheaf gluing check, not extended by a new mechanism.
3. **The obstruction is cohomological, and it unifies the two regimes (Section 7).** H^0 is the convergent states; H^1 (in the invertible fragment) is the holonomy of the morphism cycles, generated by a cycle basis. H^1 vanishes in exactly two ways, and they are the two convergence regimes: an acyclic network (no cycles) and a monotone cycle (a least fixed point). The dichotomy the base work states as two conditions is one invariant.
4. **A computable obstruction diagnostic (Section 8).** For a general (possibly non-invertible) cycle, a global convergent state exists iff the cycle’s loop composite has a reachable fixed point, and the witness for its absence is the loop composite itself, computable over the shared subspace rather than the product. This is implemented in the reference engine as a cycle diagnostic that names the offending cycle and reports whether its repair settles or orbits.
5. **Minimal coordination (Section 9).** When the obstruction is nonzero, coordinating a cycle basis of the morphism network trivializes it, yielding a convergent implementation that is strongly consistent only on those loops and coordination-free everywhere else; in the invertible fragment this core is minimal, of size the first Betti number. This refines the global coordination verdict of CALM [5, 6] and invariant confluence [7] into a localized, minimal, mixed-consistency partition, and it upgrades the engine’s blunt rejection of a nonconvergent cycle into an actionable “coordinate these loops.”
6. **Mechanization.** The structural core (Contributions 1 and 2: the limit, the retraction for arbitrary acyclic federations, and compositionality) is mechanized axiom-free in Coq/Rocq, extending the machine-checked development of the base results.

2 Related Work

Sheaf cohomology for distributed systems: the same tool, a different object. The use of sheaf cohomology to measure a local-to-global obstruction is not new, and we do not claim it. Cellular-sheaf cohomology characterizes solvability of distributed decision *tasks*, where terminating solutions are the global sections and the cohomology encodes the obstruction to a solution [9], in the combinatorial-topology lineage of distributed computing [10]; the same presheaf-and-obstruction template measures quantum contextuality [8]. Our object is different. Those works concern which one-shot decision task a set of processes can solve under a failure or adversary model. We concern the convergence of a *federated normalization*: a network of registries, each with invariants and compensation, whose morphisms must agree on shared state so that all processors reach the same valid global normal form regardless of event order. Our presheaf is over the overlaps of a normalizer structure, its sections are convergence certificates, H^0 is the convergent states, and H^1 is the holonomy of the morphism cycles. To our knowledge no prior sheaf-cohomology work targets the eventual-consistency or replicated-convergence obstruction, and none yields a coordination-minimality result.

Coordination avoidance: a global verdict we localize. The CALM theorem states that a problem has a coordination-free, consistent implementation iff it is monotone [5, 6], and invariant confluence [7] gives a related global avoidability criterion; recent work sharpens the monotonicity boundary at the semantic level [11, 12]. These answer *whether* coordination can be avoided, as a global property of the whole problem. Our minimal-coordination result answers *where* and *how little*: it locates the irreducible coordination in a cycle basis of the morphism network and shows the rest can remain coordination-free, a localized mixed-consistency partition rather than a single yes-or-no. In the information-preserving fragment the core is provably minimal. We are not aware of prior work that localizes the coordination requirement to a cycle basis or bounds it by a Betti number.

Convergent replicated data and its algebra. Conflict-free replicated data types achieve convergence by restricting to operations that never need repair; a state-based CRDT is the join-semilattice special case [4], and the monotone regime here coincides with it (this correspondence is machine-checked in the base development). The well-founded-compensation regime is strictly broader, since it admits operations that violate invariants and are repaired. Category-theoretic accounts of CRDT convergence exist at the level of the merge algebra; they do not give a cohomological obstruction or a coordination-minimality theorem. The lattice-theoretic and chaotic-iteration machinery behind the monotone regime is classical [13, 14], as is the confluence core [3].

3 Background

We restate only what later sections use: the single-registry model (well-founded compensation and compensation commutativity, giving unique normal forms via Newman’s Lemma) and the federated model (registry morphisms, the authority argument, and the single-source *M1* and multi-source *R1/R2* conditions). Full statements and proofs are in the base papers [1, 2]. [To be written: a self-contained half-page recap.]

4 Normal Forms Are a Limit

We work at the semantic level. A registry R has a finite state space D_R and a *normalizer* $\rho_R : D_R \rightarrow D_R$, the iterated-compensation map, which is total (termination, from well-founded compensation) and idempotent ($\rho_R \circ \rho_R = \rho_R$: a normal form is a fixed point, from confluence plus termination). Write $\Phi_R := \text{im}(\rho_R)$ for the valid set, the normal forms.

Everything in this section is elementary in **Set**, and the load-bearing facts are mechanized axiom-free in Coq/Rocq (module **Categorical.v**); we state them and sketch the arguments, naming the mechanized lemmas.

4.1 A single registry is an equalizer

Lemma 4.1 (Lemma 0). *For an idempotent normalizer ρ_R , the valid set, the fixed-point set, and the equalizer of the identity and ρ_R coincide:*

$$\Phi_R = \text{im}(\rho_R) = \text{Fix}(\rho_R) = \text{eq}(\text{id}_{D_R}, \rho_R),$$

a limit in Set.

Proof. Idempotence gives $\text{im}(\rho_R) = \text{Fix}(\rho_R)$: if $s = \rho_R(x)$ then $\rho_R(s) = \rho_R(\rho_R(x)) = \rho_R(x) = s$, and conversely a fixed point is its own image. And $\text{Fix}(\rho_R) = \{s : \text{id}(s) = \rho_R(s)\}$ is by definition the equalizer of id and ρ_R , a limit. Mechanized as `image_iff_fixed` and `fixed_is_equalizer`. $\square$

Read as a retraction onto Φ_R (it fixes Φ_R and lands in it), ρ_R is an idempotent monad on **Set** whose algebras are the normal forms, and a closure operator on the ordered domains of the cyclic regime. We need only the equalizer.

4.2 A federation is a finite limit

Fix a federation: registries $R_1, \dots, R_n$, and for each target B with source set $\text{src}(B)$, shared variables $\text{sh}(B)$ and a resolver Res_B merging its sources' normal forms into B 's shared component (for a single source this is the morphism's map). The base papers certify by finite enumeration that each Res_B reads only its sources ($R1$) and preserves target validity ($R2$; for a single source, $M1$). Let g send a tuple of component normal forms to the tuple of the targets' current shared components, and h send it to the tuple of resolver values. The *consistent set* is

$$L_F := \left\{ (s_1, \dots, s_n) \in \prod_i \Phi_{R_i} : \text{for every target } B, s_B \upharpoonright \text{sh}(B) = \text{Res}_B((s_A)_{A \in \text{src}(B)}) \right\}.$$

Proposition 4.2 (Proposition 1). $L_F = \text{eq}(g, h)$. *Since each Φ_{R_i} is a limit (Lemma 4.1) and L_F is an equalizer of a product of these, L_F is a finite limit in **Set**.*

Proof. A tuple lies in L_F iff every target's shared component equals its resolver value, that is, iff g and h agree on it componentwise, that is, iff $g = h$ there. Componentwise agreement over the finite target set is equality of the two tuple-valued maps, so L_F is their equalizer. Mechanized as `consistent_iff_equalizer`, with the product over targets modeled as a list so the characterization needs no functional extensionality. $\square$

4.3 The federated normalizer is the retraction onto the limit

Fix a topological order (sources before targets). The federated normalizer ρ_F processes registries in that order: for each, overwrite its shared component with the resolver applied to the already-finalized sources, then apply the local normalizer. We model this as a left-to-right fold over the ordered components, writing `step` for the per-position operation (read finalized prefix, overwrite shared, normalize). The one hypothesis it needs is that `step` is idempotent given a fixed finalized prefix, which is exactly $M1/R2$ (the local normalizer fixes the overwritten shared component) together with local idempotence.

Theorem 4.3 (Theorem 1: federation retraction, acyclic). ρ_F is an idempotent retraction onto L_F : $\text{im}(\rho_F) = \text{Fix}(\rho_F) = L_F$. *Equivalently, the federated normal forms are exactly the finite limit of Proposition 4.2 and ρ_F is the retraction onto it. Moreover ρ_F is independent of the chosen topological order.*

Proof sketch. Two facts suffice and yield the retraction by a general argument: any operator that is sound and complete for a set L is the idempotent retraction onto L , with image and fixed set L (mechanized as `RetractionOntoConsistent`). *Soundness* (Lemma A): $\rho_F(s) \in L_F$, because each finalized position is step of its finalized prefix, which step-idempotence leaves fixed (`rhoFold_sound`). *Completeness* (Lemma B): a consistent state is already fixed,

since each position's step is a no-op on it; this uses only the definition of L_F , not $M1/R2$ (`rhoFold_complete`). Hence ρ_F is the idempotent retraction onto L_F (`rhoFold_retraction`, `rhoFold_image_iff_consistent`), for arbitrary acyclic federations, not a fixed small instance.

Order-independence reduces to the commutation of two incomparable registries: their reads and writes are disjoint, so processing them in either order gives the same result (`updates_commute`). Lifting this to all topological orders uses the classical fact that the linear extensions of a finite poset are connected by adjacent transpositions of incomparable elements, which we take as given rather than mechanize. $\square$

5 Compositionality for Free

Because the federated normal forms are a limit and ρ_F is the retraction onto it, a convergent sub-federation behaves as a single effective registry inside a larger one. Split a federation along a topological cut into an upstream block J (closed under sources: every source of a J -registry is itself in J) and the rest K . The *flat* normalizer runs ρ_F over all of $J \cup K$ in one order; the *staged* normalizer runs the sub-federation's normalizer ρ_J to completion, then runs K over the finalized J -block.

Theorem 5.1 (Theorem 2: compositionality). *The flat and staged normalizers agree: normalizing $J \cup K$ in one pass equals normalizing J , then normalizing K over the finalized result. Consequently a verified sub-federation collapses to an effective registry (its valid set L_J), and the combined normalizer factors through it.*

Proof sketch. At the operator level this is the fold-append law: folding J and then continuing with K from the accumulated result equals folding the concatenation, because the accumulator after J is exactly ρ_J 's output and each K -step reads only its own finalized sources. Mechanized as `rhoFold_compositional`, via `rhoF_from_app`, axiom-free. $\square$

This is what lets verification scale: a subsystem is checked once, in isolation, and reused as a black box, with only the seam (the morphisms crossing the cut into K) re-checked. The assume-guarantee refinement makes the black box parametric. A sub-federation may declare *input ports*: shared variables that no internal morphism writes, which an outer morphism may drive once the subsystem is embedded. A seam morphism into a port is verified against $M1/R2$ while the subsystem's interior is trusted from its certificate, and a write to any non-port (sealed) variable is rejected. We treat the port refinement at the paper level here; the operator-level collapse above is the mechanized core.

6 Convergence Certificates Form a Sheaf

[To be written from CATEGORICAL-STRUCTURE.md §10. The presheaf of certificates, the non-naive gluing counterexample, the precise gluing condition (agree as normalizers on the overlap), and the identification of $R1 + R2$ with the gluing axiom.]

7 The Cohomological Completion

[To be written from CATEGORICAL-STRUCTURE.md §10.1. H^0 as the convergent states; H^1 as the holonomy in the invertible fragment; the completion theorem; the cycle-basis generators;

the two routes to $H^1 = 0$ recovering the acyclic and monotone regimes; and the reduction of the general non-invertible case to the loop-composite fixed-point condition.]

8 The Obstruction Diagnostic

[To be written from CATEGORICAL-STRUCTURE.md §10. The loop composite as the computable witness, bounded by the shared subspace; the settle-versus-orbit dichotomy; and the reference implementation (`Federation.DiagnoseCycle`), with worked settle and orbit examples.]

9 Minimal Coordination

[To be written from CATEGORICAL-STRUCTURE.md §10.2, incorporating the inhabitation analysis. The minimal-coordination proposition; the coordination-free-majority plus coordinated-core decomposition; the characterization of the invertible (lossless-relabeling) fragment and which real federations inhabit it; the accept-with-coordination capability; and the positioning against CALM and invariant confluence.]

10 Conclusion

[To be written.]

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